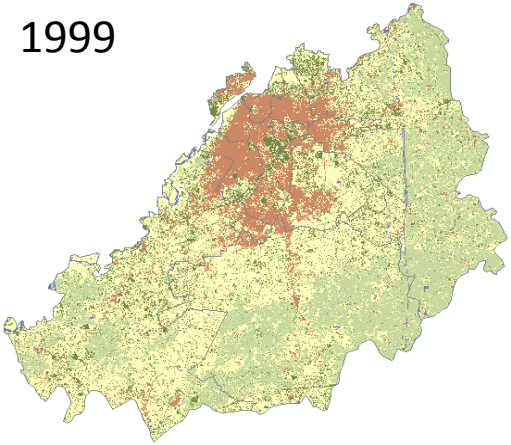
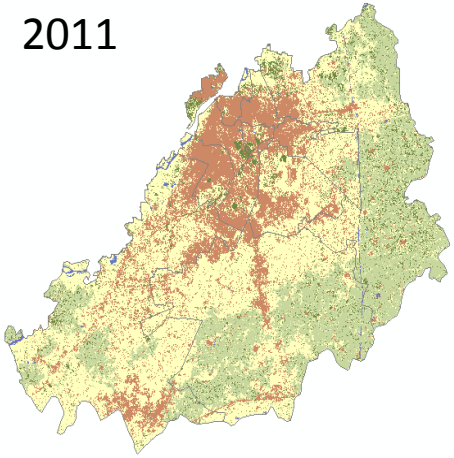
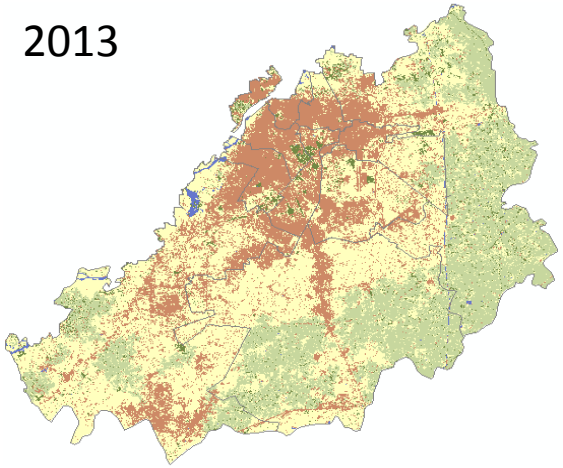
1999



2011

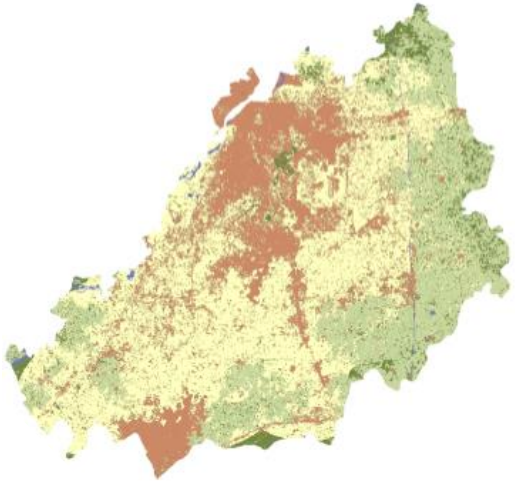


2013

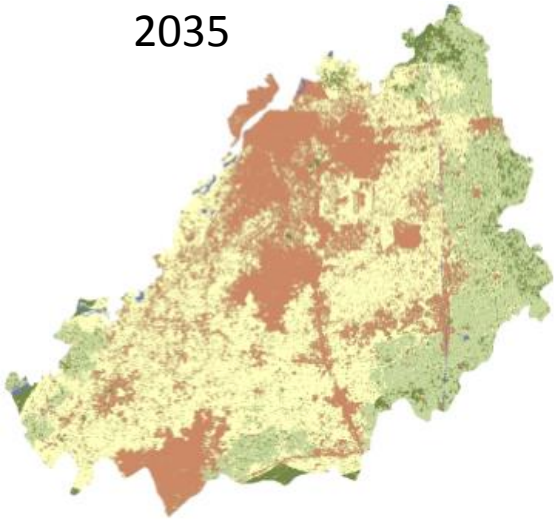


LAND USE MODELS

2021



2035

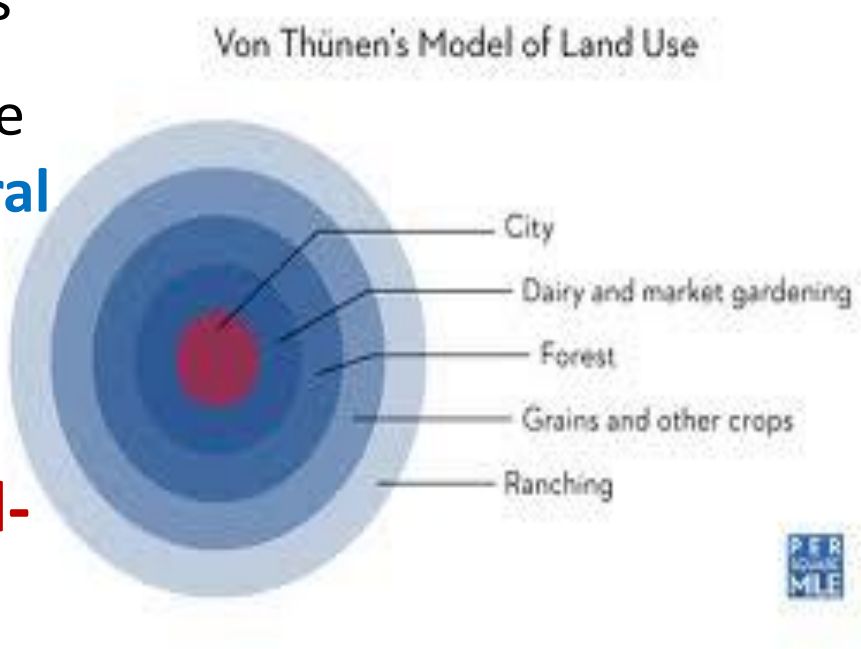


Introduction

- Land use planning is that part of the process of city planning concerned with the type, location, intensity, and amount of land development required for the various space-using functions of urban life.
- Determining the proper relationships of various land uses to one another and to existing and proposed transportation facilities, utilities, and community facilities.
- The land use plan is usually expressed in a colored plan map, supported by pertinent text, tables, and graphs setting forth recommended standards

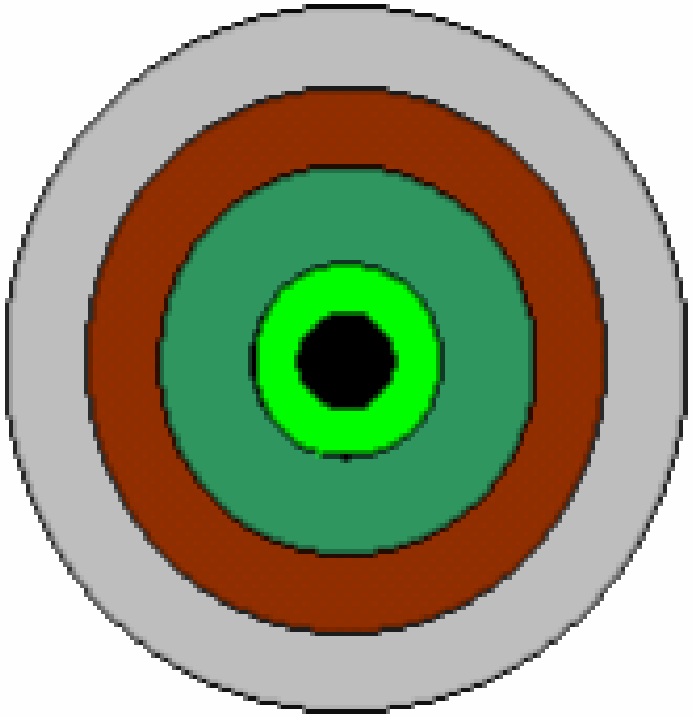
Von Thünen model

- **Von Thünen' Agricultural Land Use Theory: in 1826**
  - A German skilled farmer who was knowledgeable in economics
  - To explain the patterns of the distribution of the **agricultural land use and land value**
  - Ring-type land use structure
- **William Alonso's Urban Bid-Rent Theory**



Von Thünen model

- If all conditions stand true, Von Thünen hypothesized that the following pattern would develop

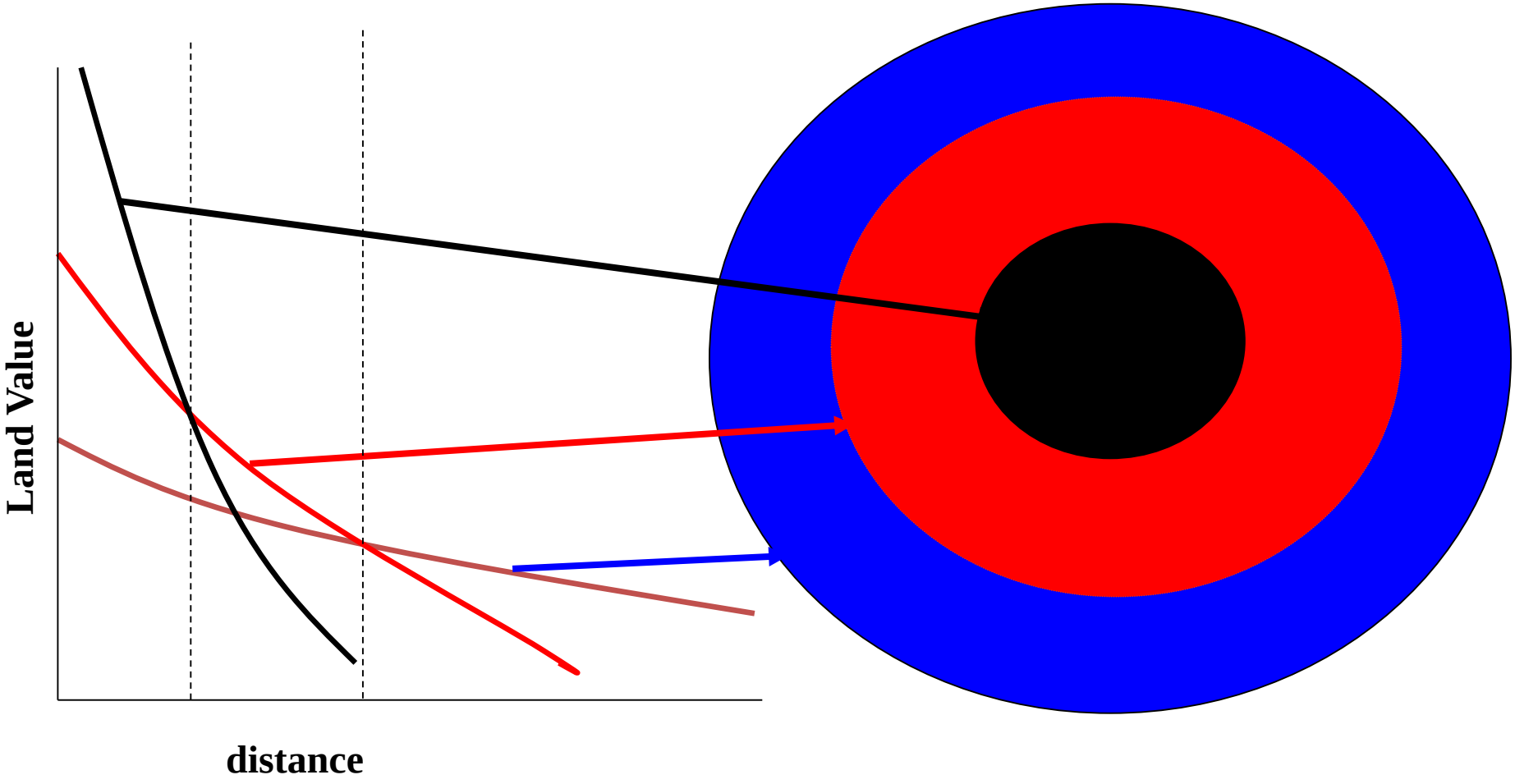




Von Thünen model

- Von Thünen's model was only **concerned** with **location**.
- We can derive the land rent from this model.
  - **closer to the city, higher will be the profit**
- **Land Rent**, is at most the profit they make at that location

Von Thünen model



Von Thünen model

R = Y * (P-Cp) – Y * (Ct * d), in which:

R = land rent ($/acre)

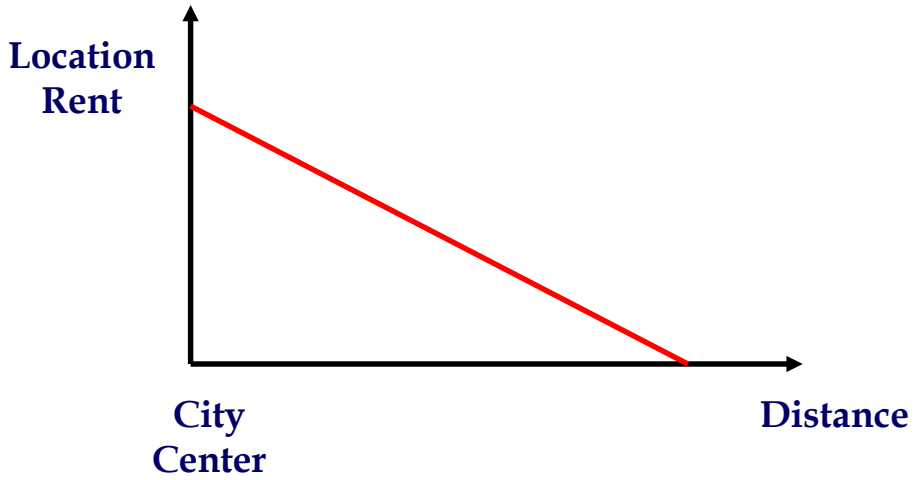
Y = yield (ton/acre)

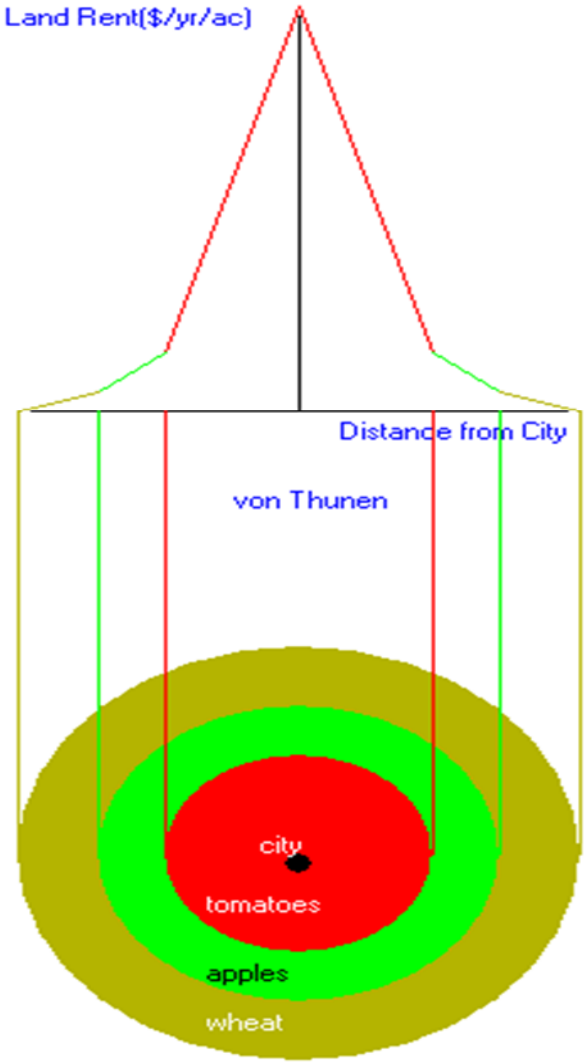
P = market price ($/ton)

Cp = production cost ($/ton)

Ct = transportation cost ($/ton/miles)

d = distance from town (miles)





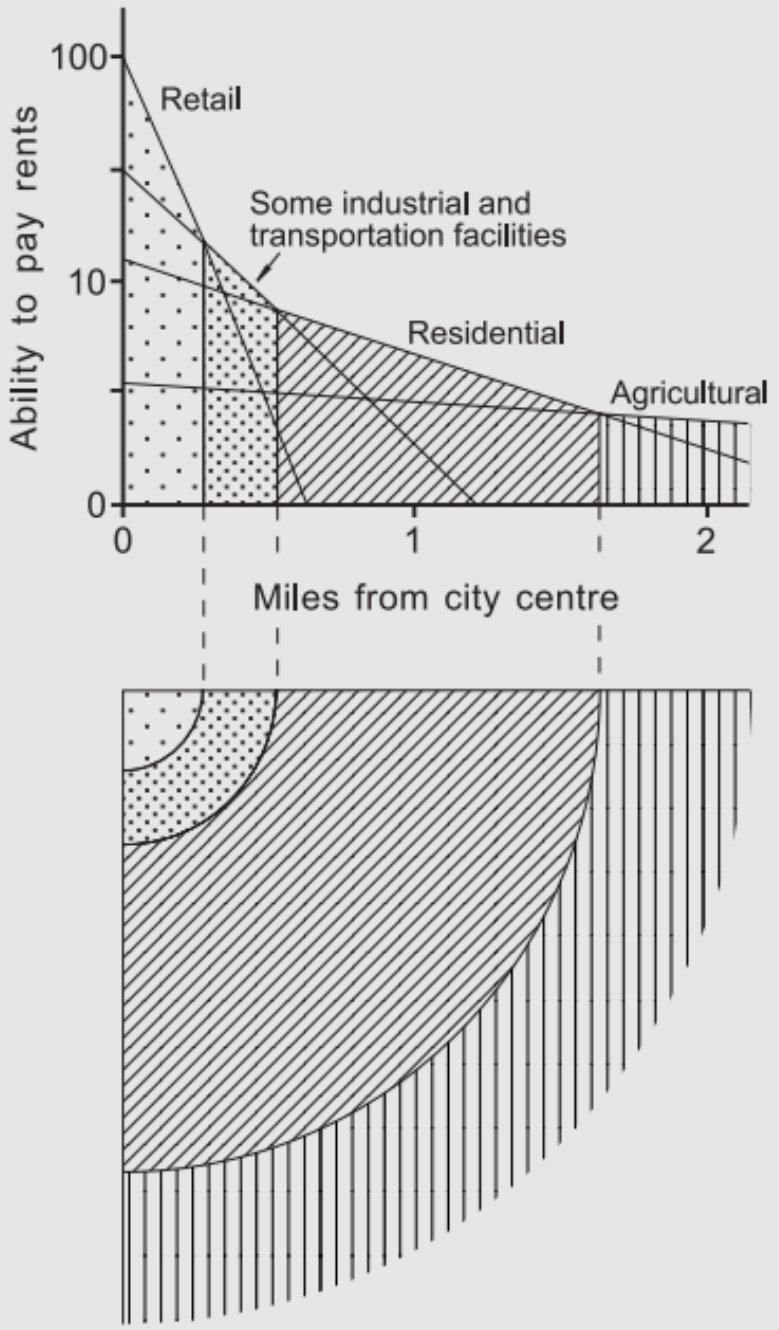
Assumptions

- The central city is located centrally within an "Isolated State" which is self-sufficient and has no external influences.
- The Isolated State is surrounded by an unoccupied wilderness.
- The land of the State is completely flat and has no rivers or mountains to interrupt the terrain.
- The soil quality and climate are consistent throughout the State.
- Farmers in the Isolated State transport their own goods to market via oxcart, across land, directly to the central city. Transport costs thus rise linearly with distance.
- The selling price for the agricultural products is determined at the market by supply and demand.
- Farmers act to maximize profits.

Bid rent theory

- Geographical economic theory that refers to how the price and demand for real estate change as the distance from the central business district (CBD)
- This is based upon the idea that retail establishments wish to maximize their profitability, so they are much more willing to pay more for land close to the CBD and less for land further away from this area.
- The amount they are willing to pay is called "bid rent".

The trade-off model of urban land use

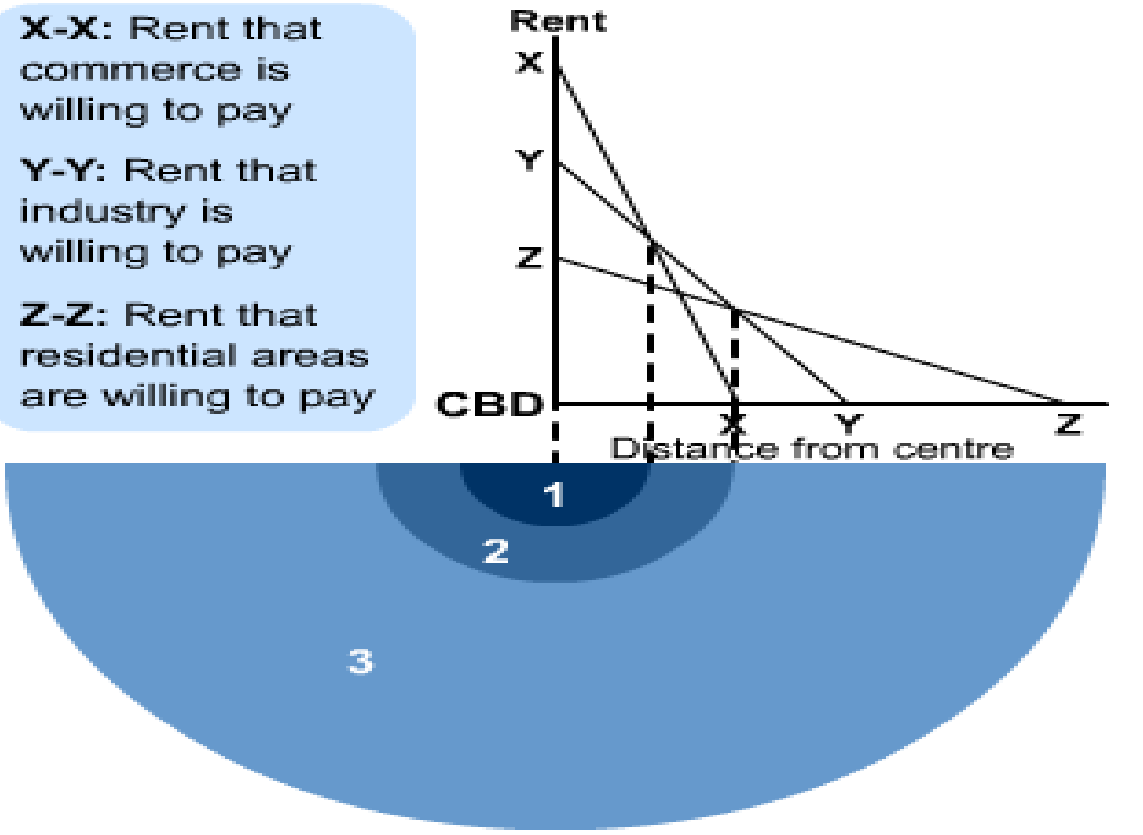


Bid rent theory

X-X: Rent that commerce is willing to pay

Y-Y: Rent that industry is willing to pay

Z-Z: Rent that residential areas are willing to pay



1: CBD with commerce and offices
2: Industry
3: Residential with highest density nearest centre

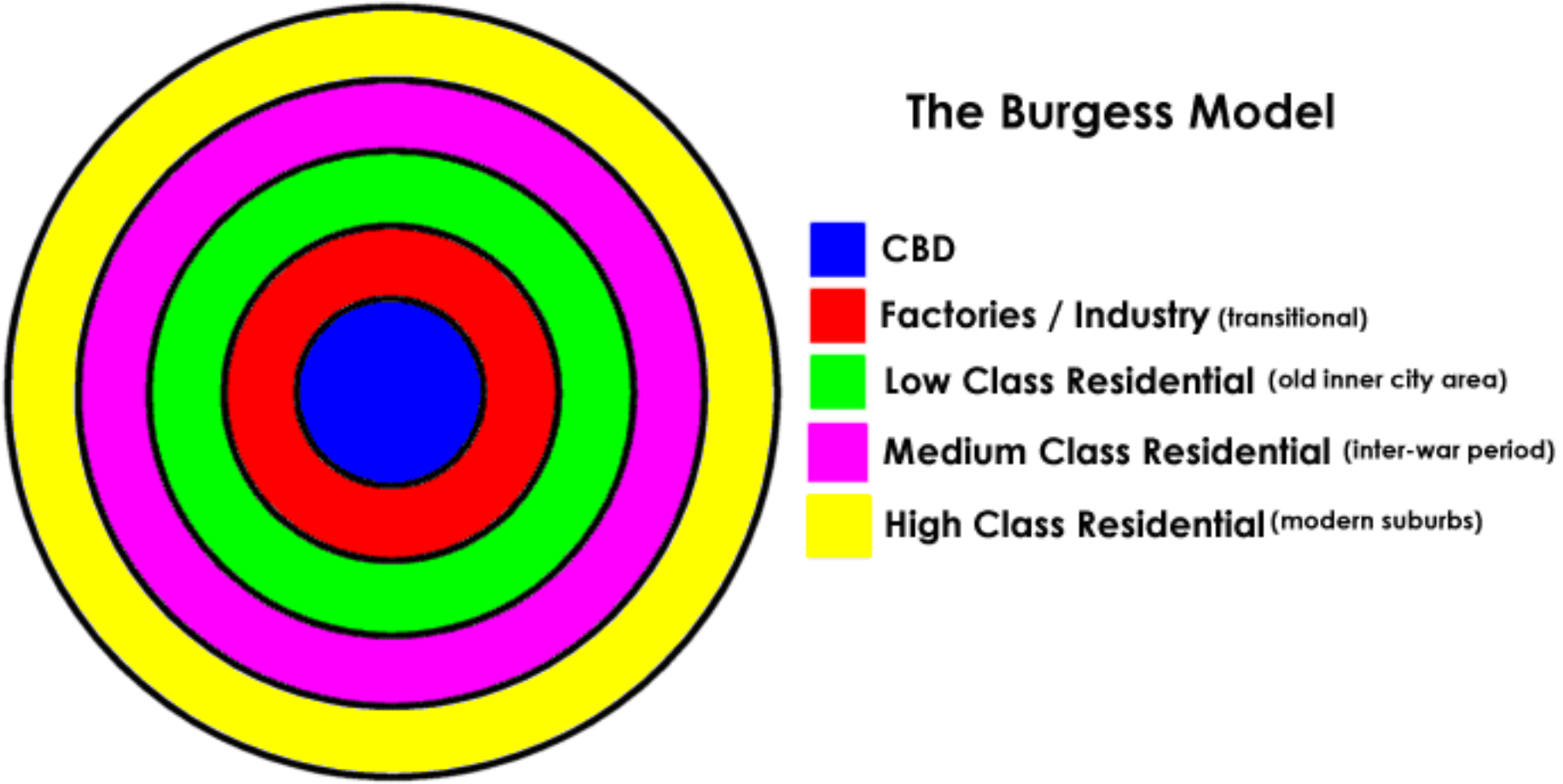
Urban land use models

- Common starting point is the CBD.
- Models differ in how they explain the location and development of the city outside of the CBD.
- Three main types of models:
  1. Concentric zone model (*Burgess* 1920's)
  2. Sector model (*Hoyt* 1930's)
  3. Multiple-nuclei model (*Harris and Ullman* 1940's)

NORTH AMERICAN MODELS

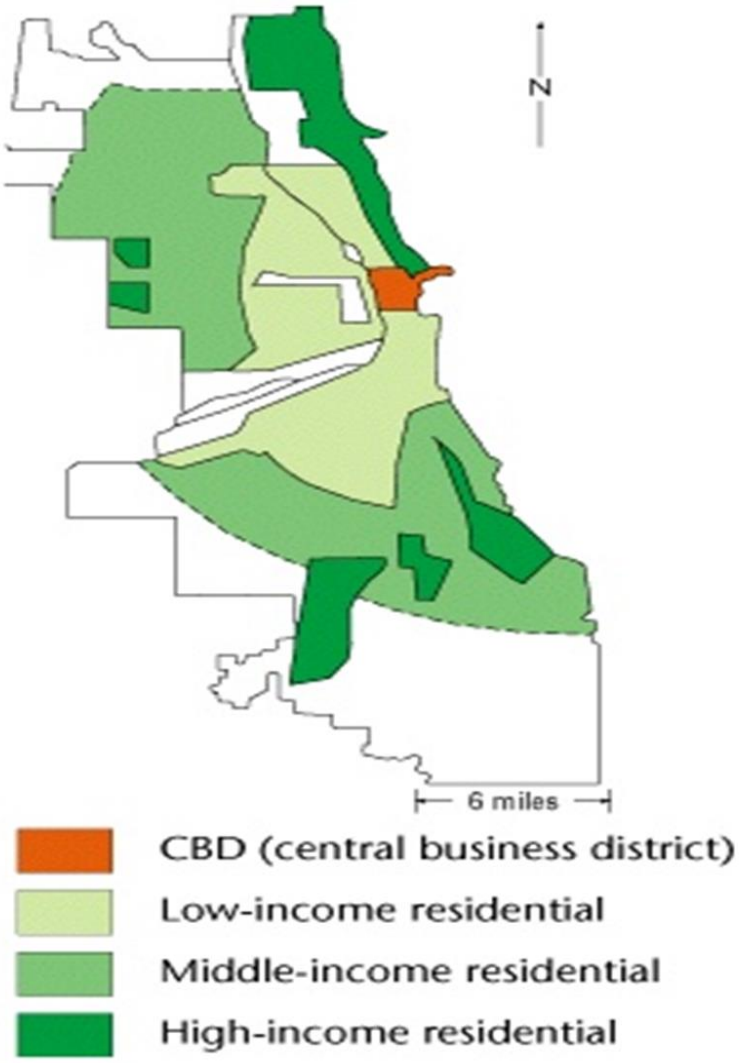
The Burgess Model

Ernest Burgess (1923)



Concentric Zone Model

- The model was produced in the 1920's much has changed since then.
- Chicago is near lake Michigan thereby making it semi-circular.
- The model was only based on Chicago – so only shows only one city at one point in time.



Concentric Zone Model

- The burgess model = the bull's eye model = the concentric ring model

- The model -Based on human ecology theories done by Burgess and applied on Chicago, it was the first to give the *explanation of distribution of social groups* within urban areas.

- In 1925, Burgess presented an urban land use model, which divided cities in a set of concentric circles expanding from the downtown to the suburbs.

Concentric Zone Model

- According to this model, a large city is divided in concentric zones with a tendency of each inner zone to expand in the other zone.

- Urban growth is thus a process of expansion and reconversion of land uses.

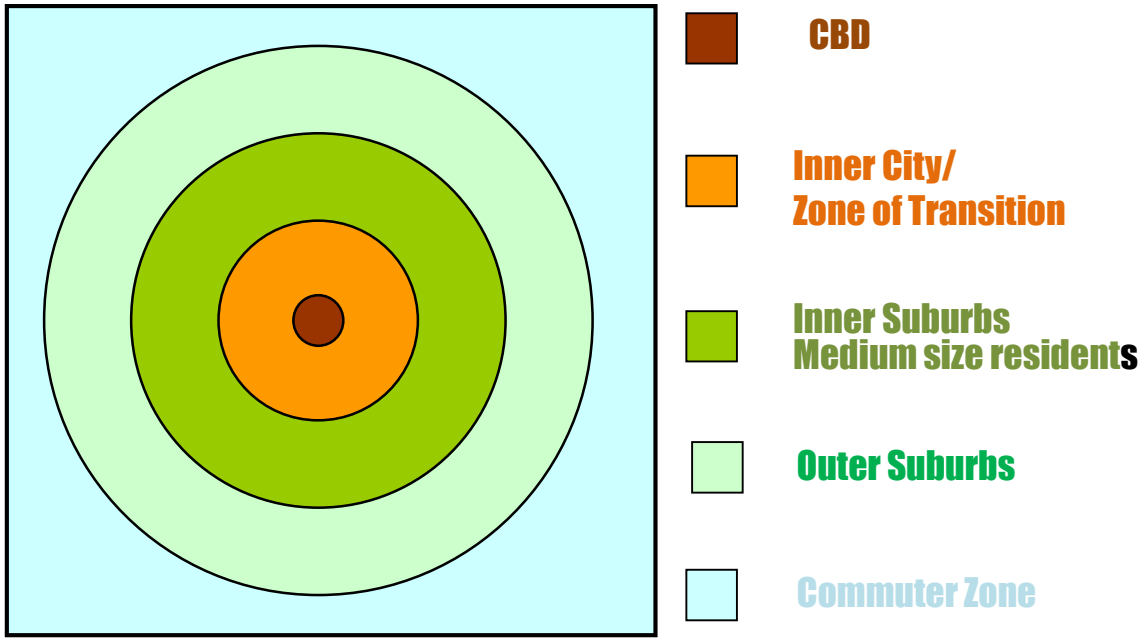
- This representation was built from Burgess's observations of a number of American cities, notably Chicago.

Concentric Zone model

1. **Central Business District (CBD)** - This area of the city is a non-residential area and it's where businesses are. This area s called downtown ,a lot of sky scrapers houses government institutions, businesses, stadiums, and restaurants
2. **Zone of Transition-** the zone of transition contains industry and has poorer-quality housing available. Created by subdividing larger houses into apartments

Concentric Zone Model

• Chicago, 1920s



1. Why are 'high buildings' found in the CBD?
2. What are the strengths and weaknesses of this model?

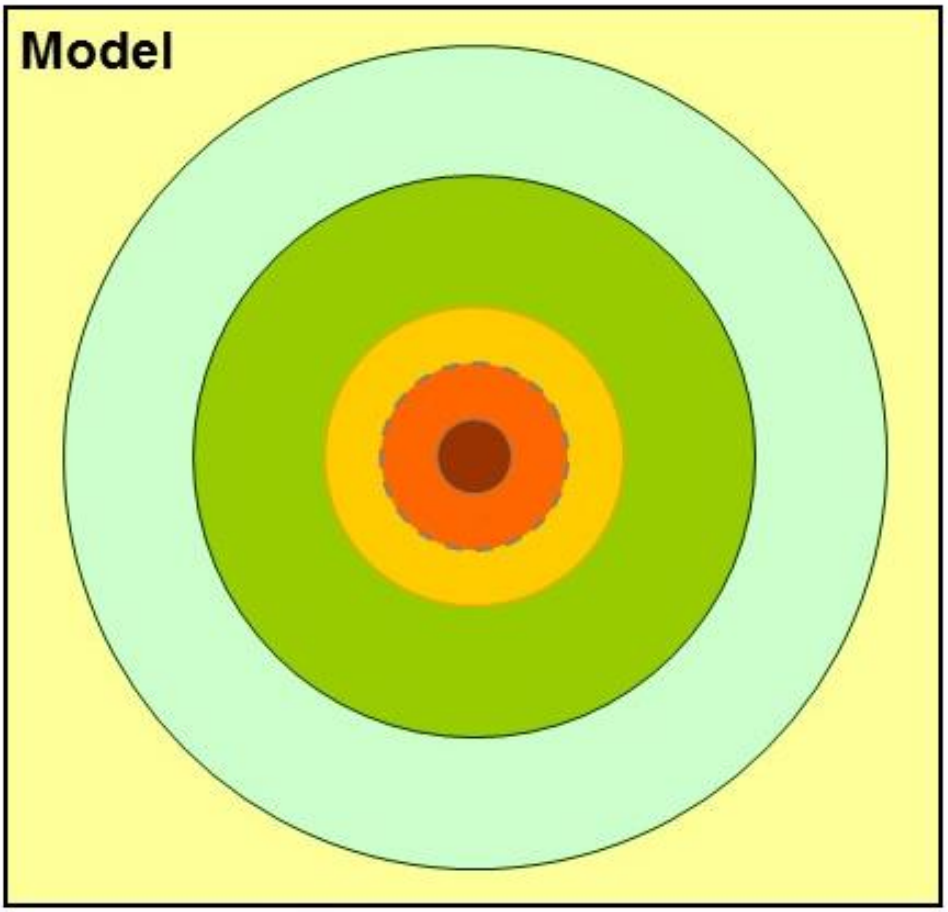
• **CBD** – core of the city. Contains shops, offices and entertainment. Few residential. High buildings.

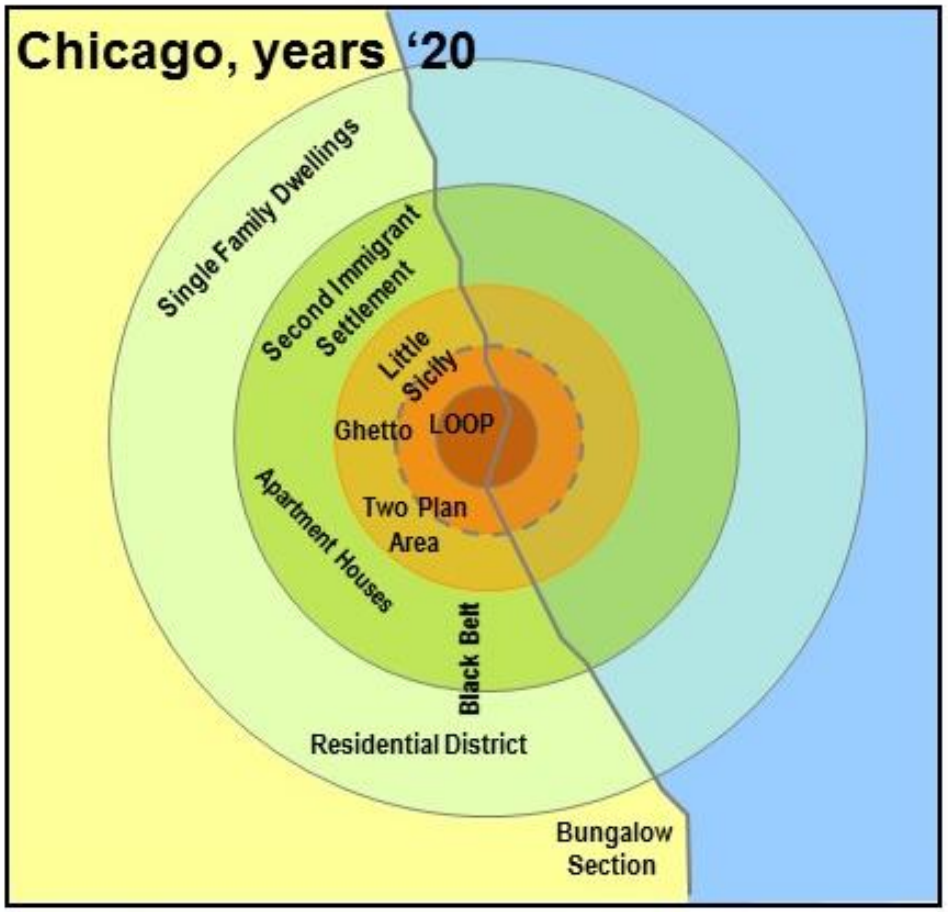
• **Inner City** – Mixed land use of small industries, small houses and offices.

• **Inner Suburbs** – Housing dates 80 – 100 yrs. Terraced houses with back yards.

• **Outer Suburbs** – Semi-detached houses 1930+ Council houses. Shopping parades. Out-of-town shops.

• **Commuter Zone** – Green belt around urban areas. Beyond it, small towns and villages. High cost housing.





I - Loop (downtown; CBD)

II – industries

III – transition area

IV – working class area

V – residential area

VI – suburban area

Concentric Zone model

3. **Zone of the working class-** This area contains modest older houses occupied by stable, working class families. A large percentage of the people in this area rent.
4. **Zone of better residence-** This zone contains newer and more spacious houses. Mostly families in the middle-class live in this zone.
5. **Commuter's Zone/Suburbs-** This area is located beyond the build-up area of the city. Mostly upper class residents live in this section.

Concentric Zone Model

Central Business District

- Major shops and Offices
- The most accessible point of the city
- The original core of the city
- Very low residential population.

Zone of transition

- Made up of old housing and factories.
- Stages of deterioration and redevelopment.
- Areas of low class housing.
- poorest people, blue collar used to live here
- Terraced or tenements.
- Over crowded
- Close to industry

Medium class residential

- Housing built during inter war period.
- Population density is lower than previous zone.
- apartments, blue collar workers (walk to work)

High Class Residential or Modern Suburbia

- Houses mainly semi detached or detached
- Well planned
- Lots of open space
- Good transport links
- Lower population density

Concentric Zone Model

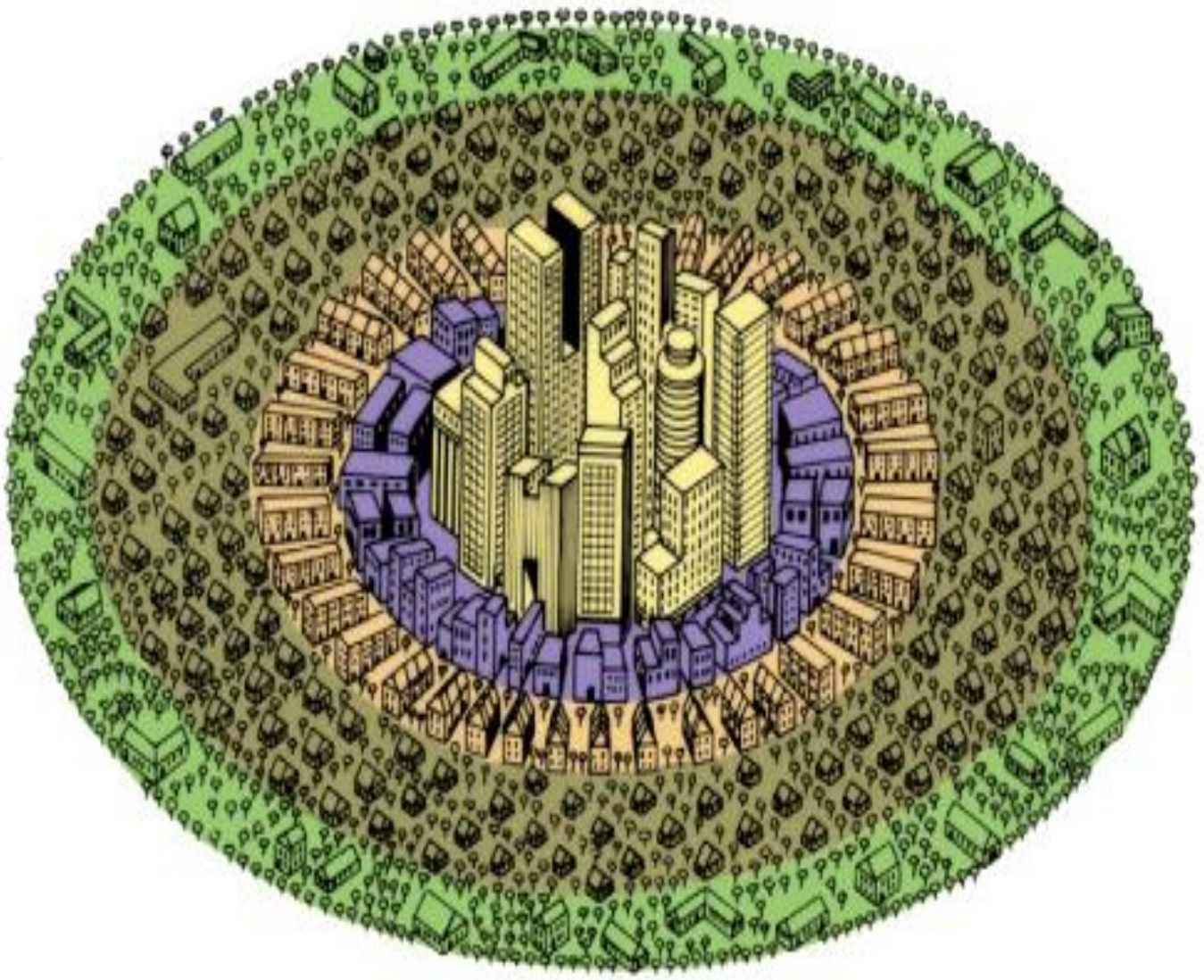
ASSUMPTIONS

- City built on a flat surface.
- Transport easy in all directions.
- Land value highest in city centre
- Oldest buildings close to city centre
- Poorer classes have to live near work and city centre.

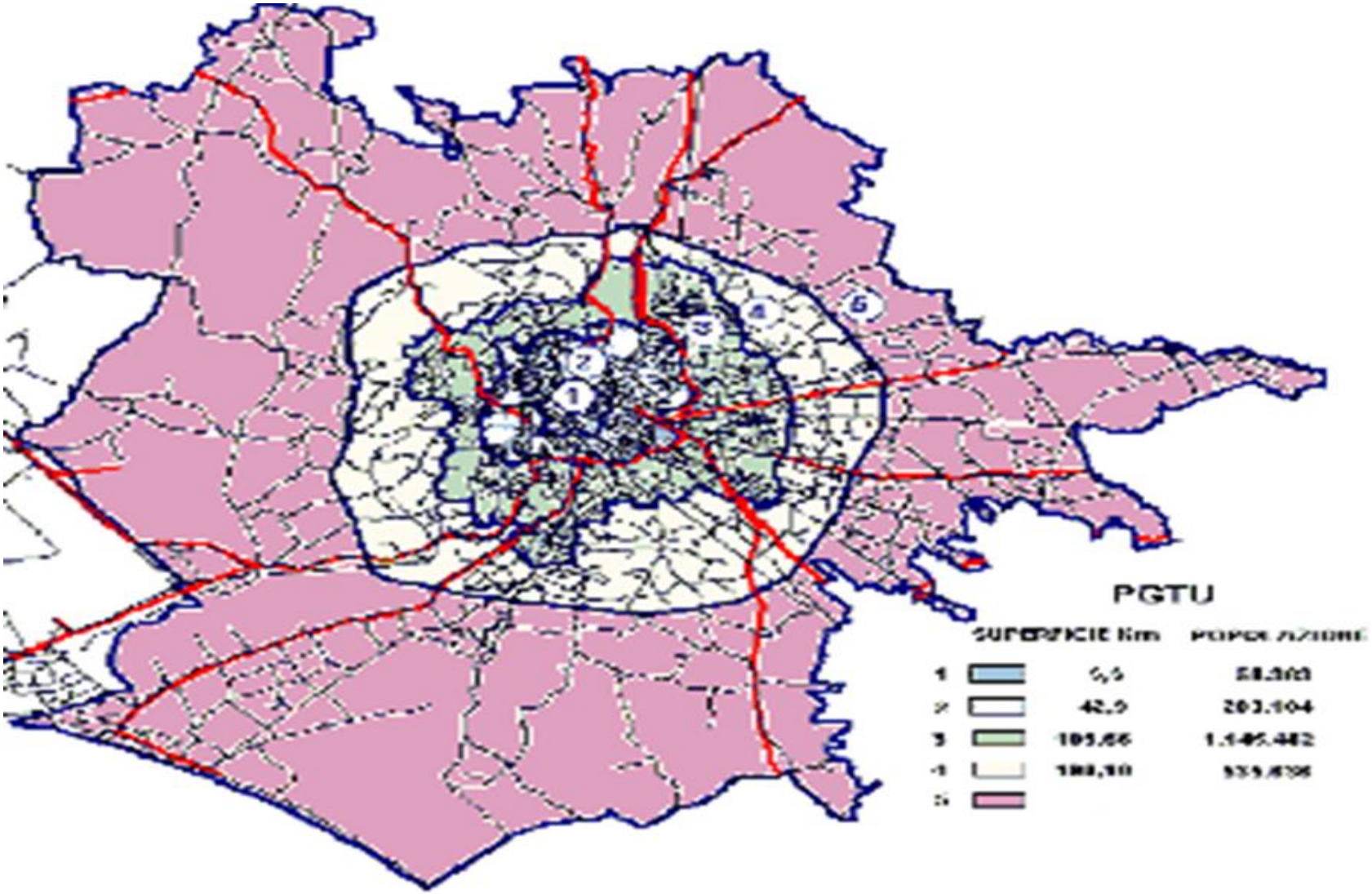
CRITICISM

- Zones are never that clearly defined.
- Zones usually have a mix of land uses
- Cities rarely on large flat plains.
- No edge of city developments.
- Oldest housing not always found in centre.
- The model was developed for a specific era in the US

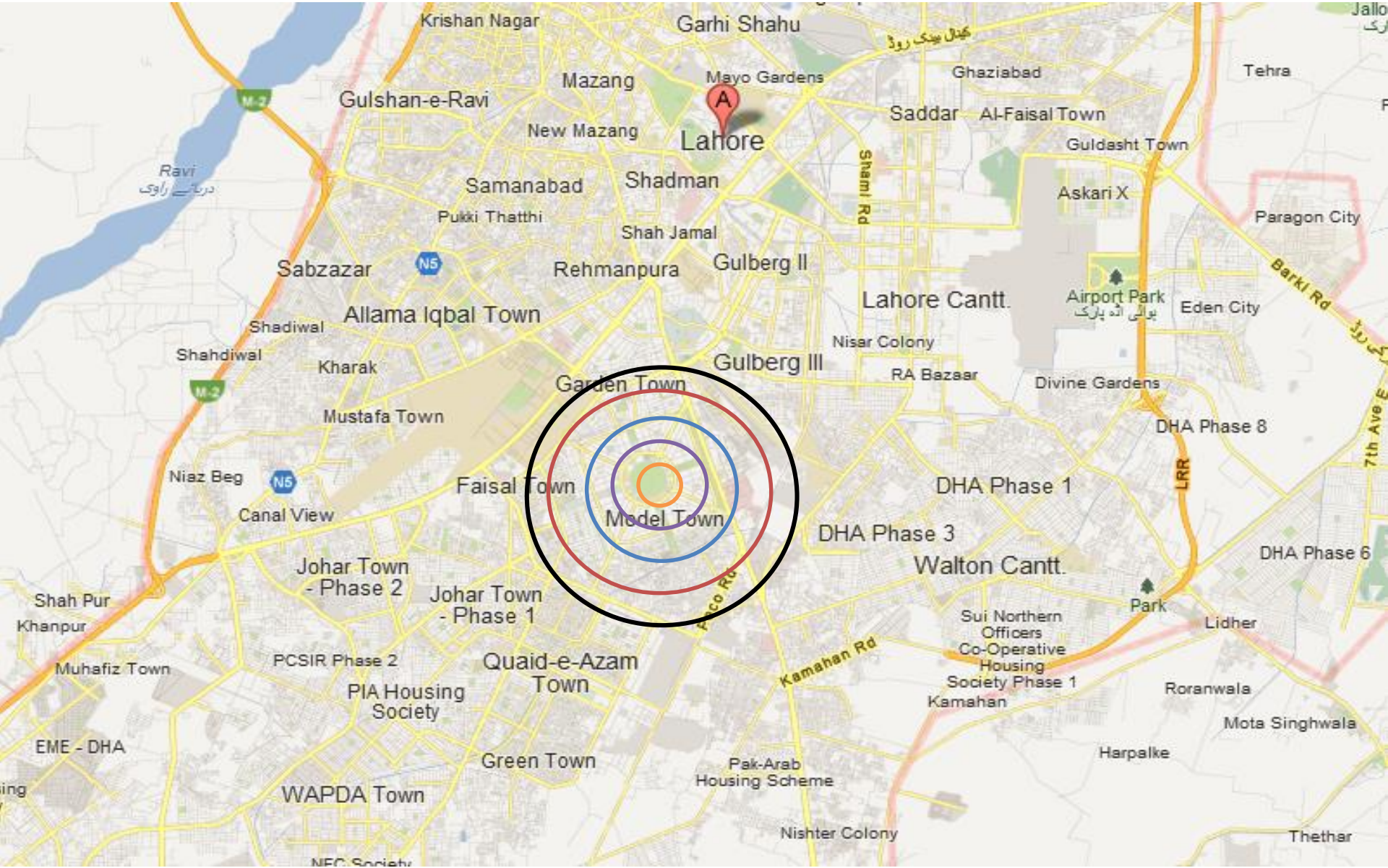
Concentric Zone Model



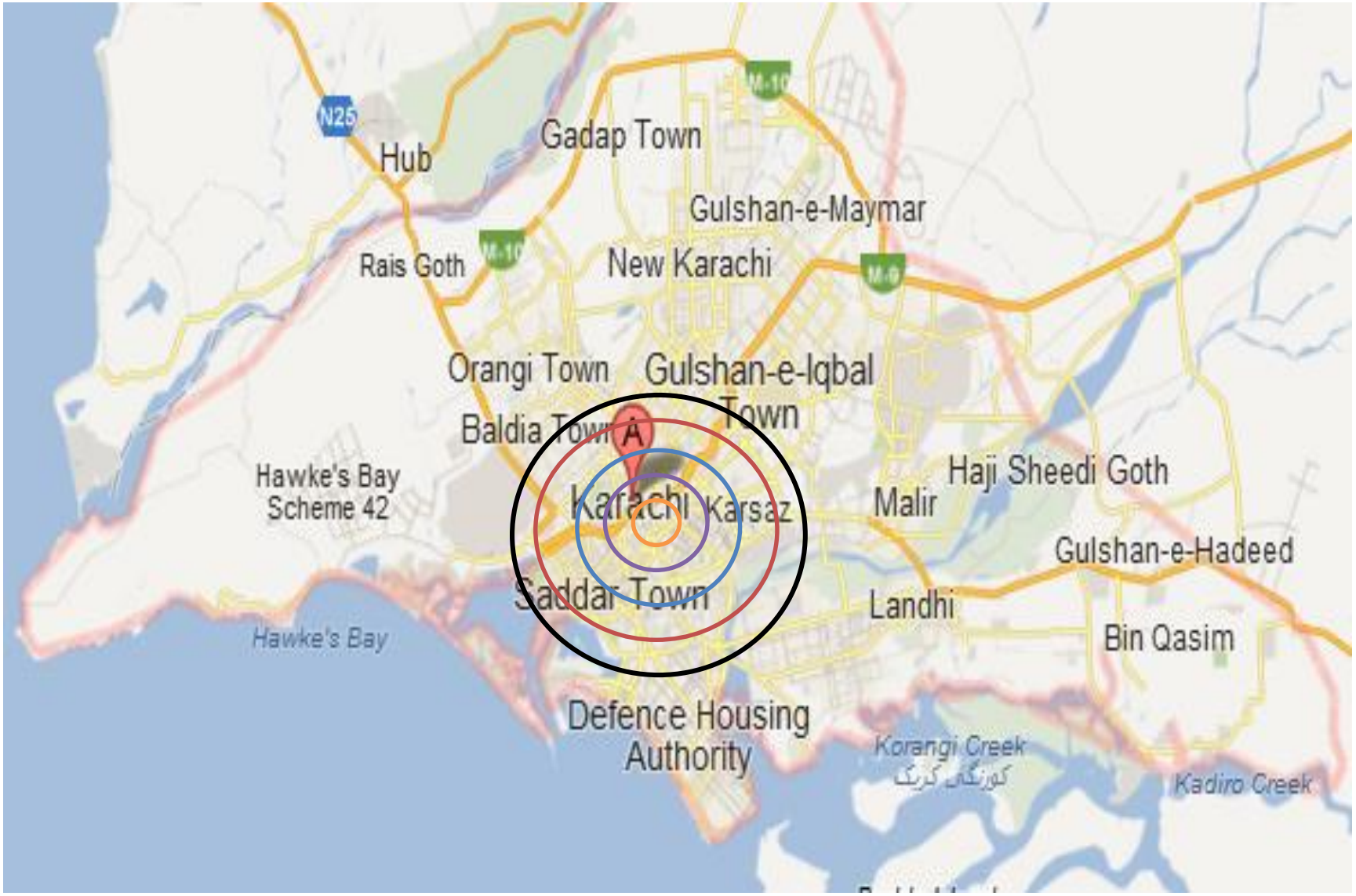
Rome: Example of Concentric zone



Concentric Zone Model applied to Lahore

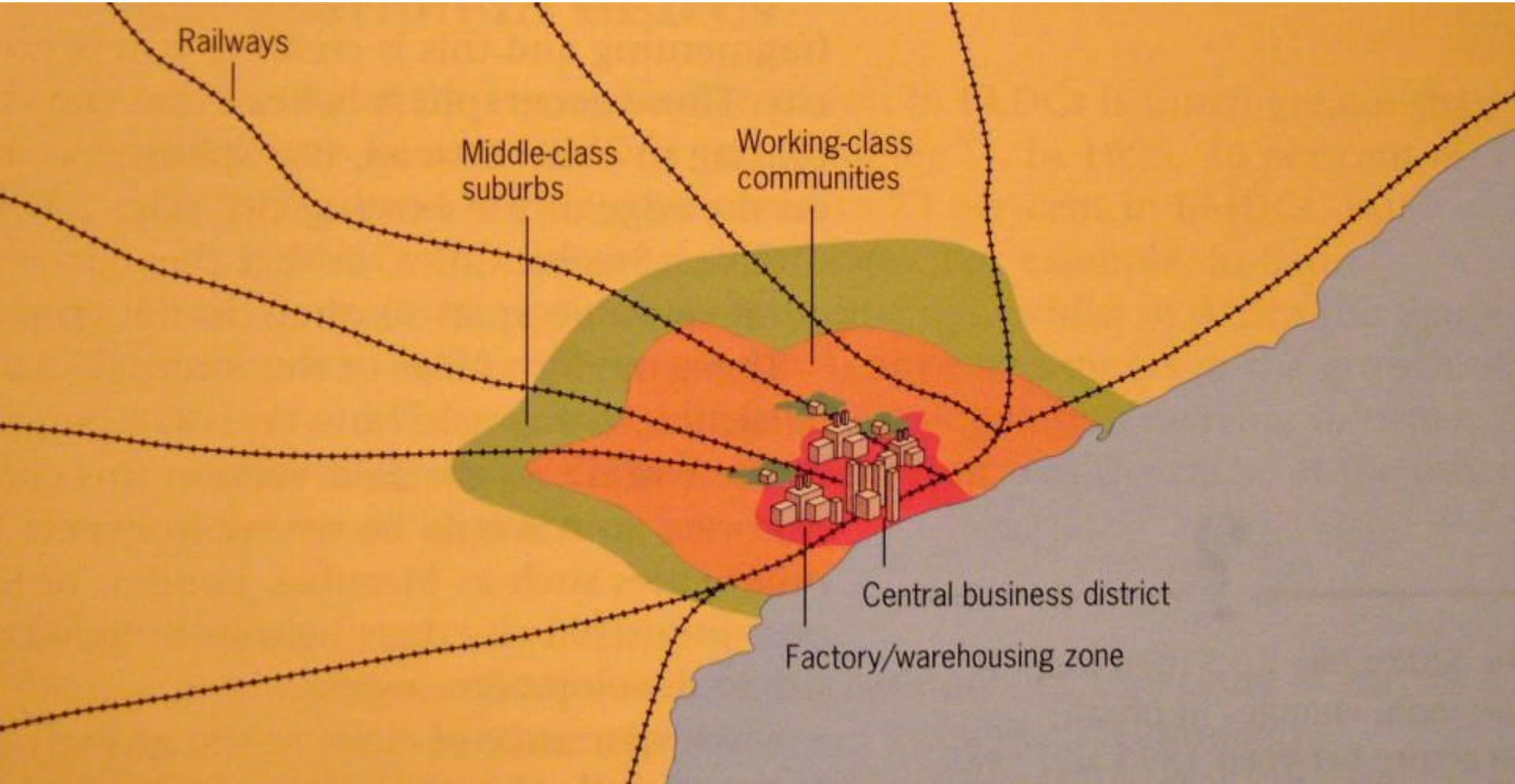


Concentric Zone Model applied to Karachi

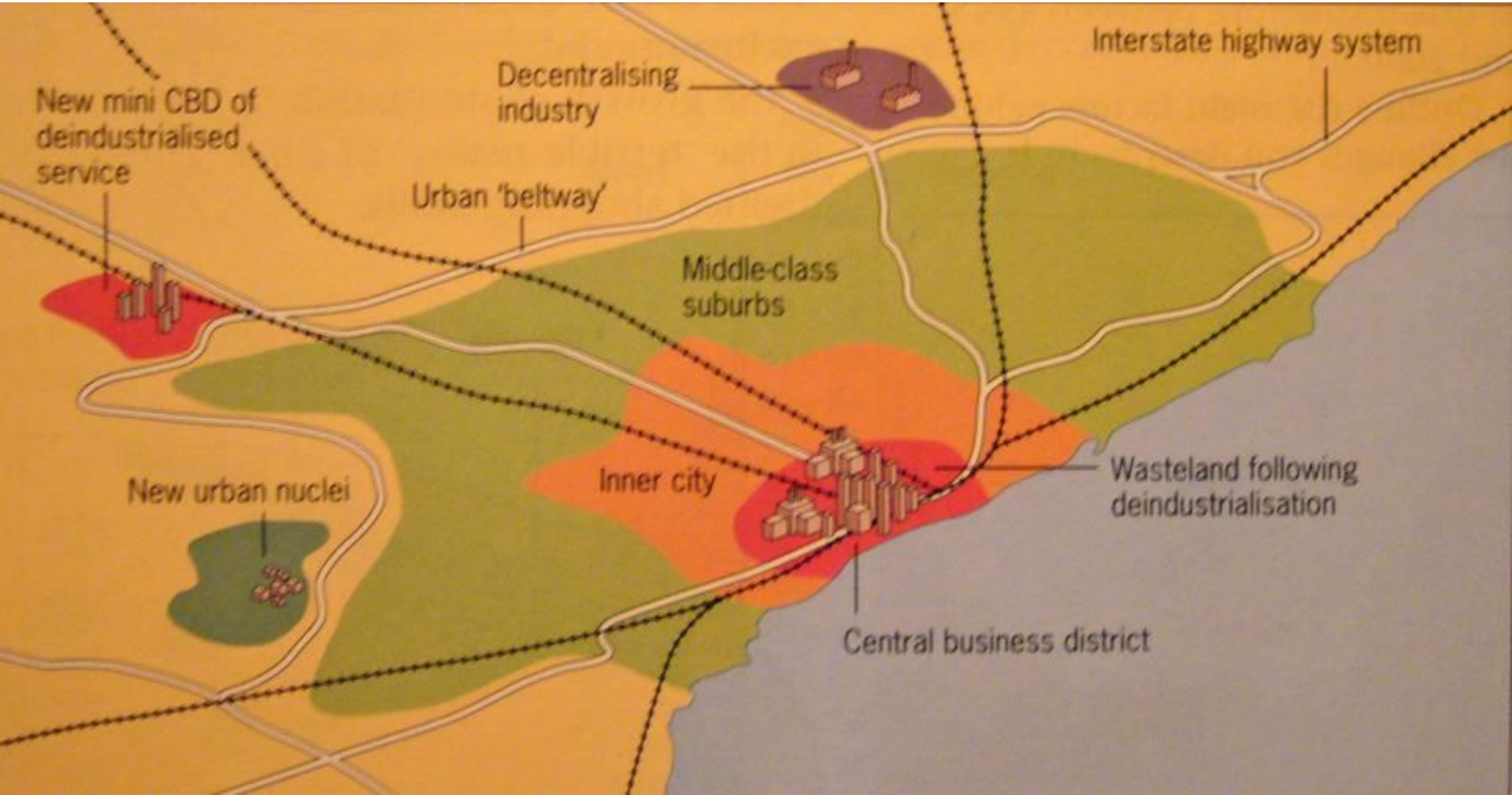


Classic industrial city 1850s

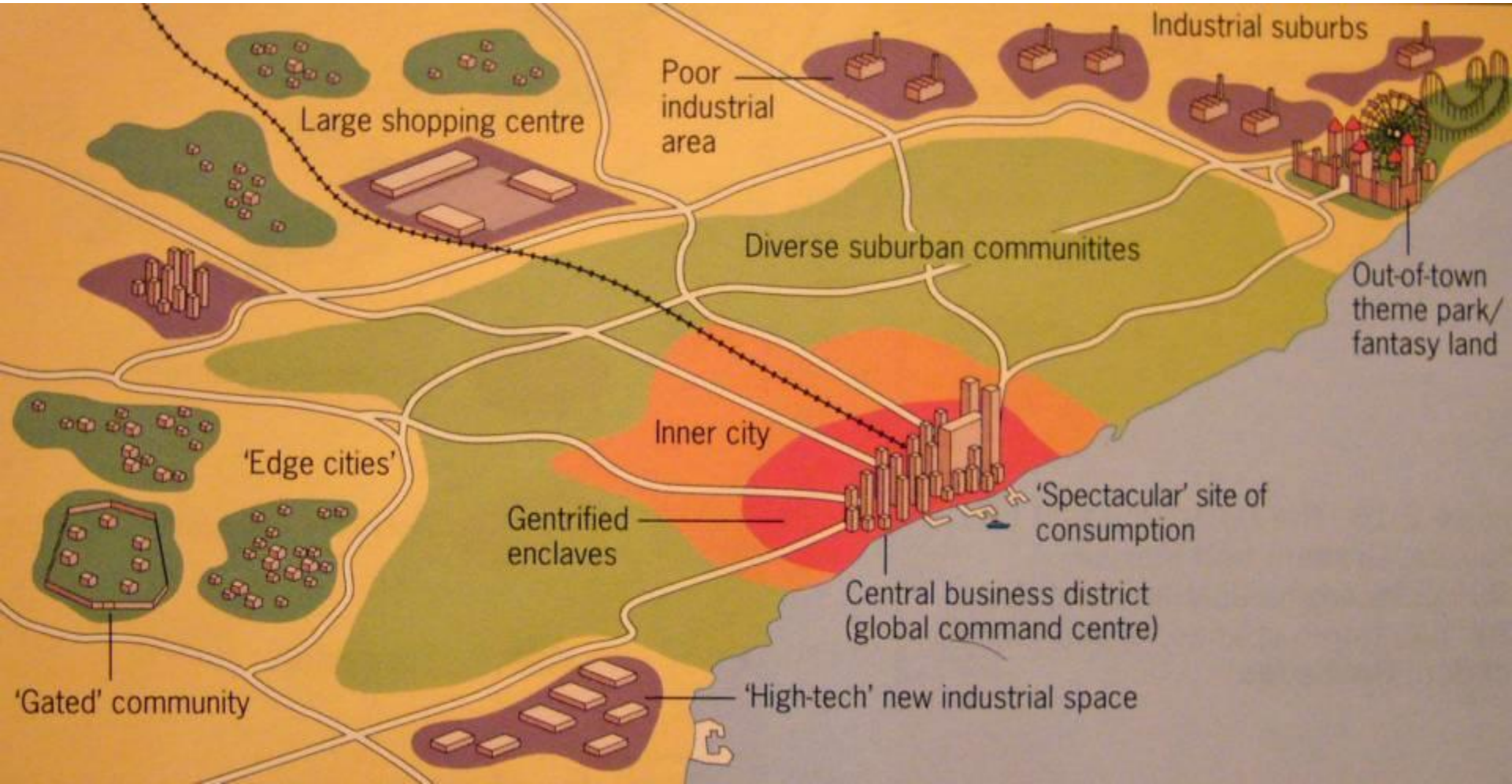
Konx and Pinch 2000 – Urban Social Geography



Industrial city 1945-1975



The post-industrial city



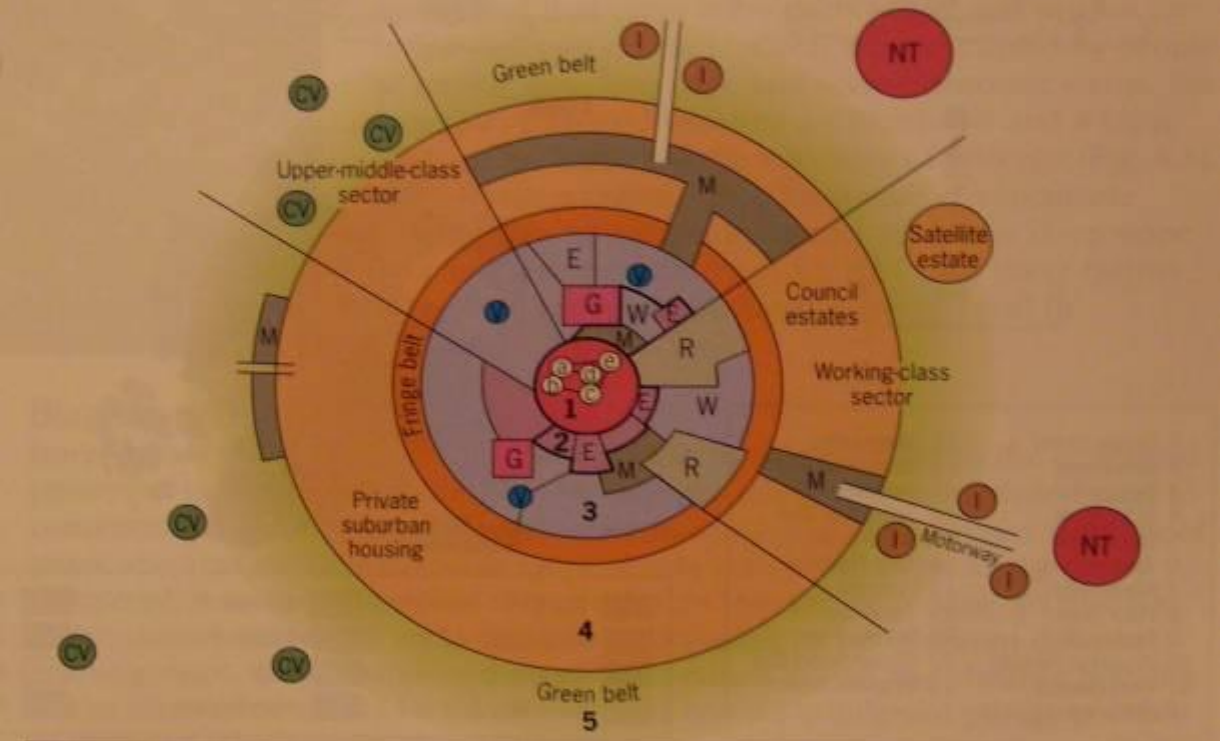
Concentric Zone model

Shortcomings

- It assumes an isotropic plain
- land may restrict growth of certain sectors
- The model does not fit polycentric cities
- It describes the peculiar American geography, where the inner city is poor while suburbs are wealthy; the converse is the norm elsewhere.

Kearsley's model of urban structure, 1983

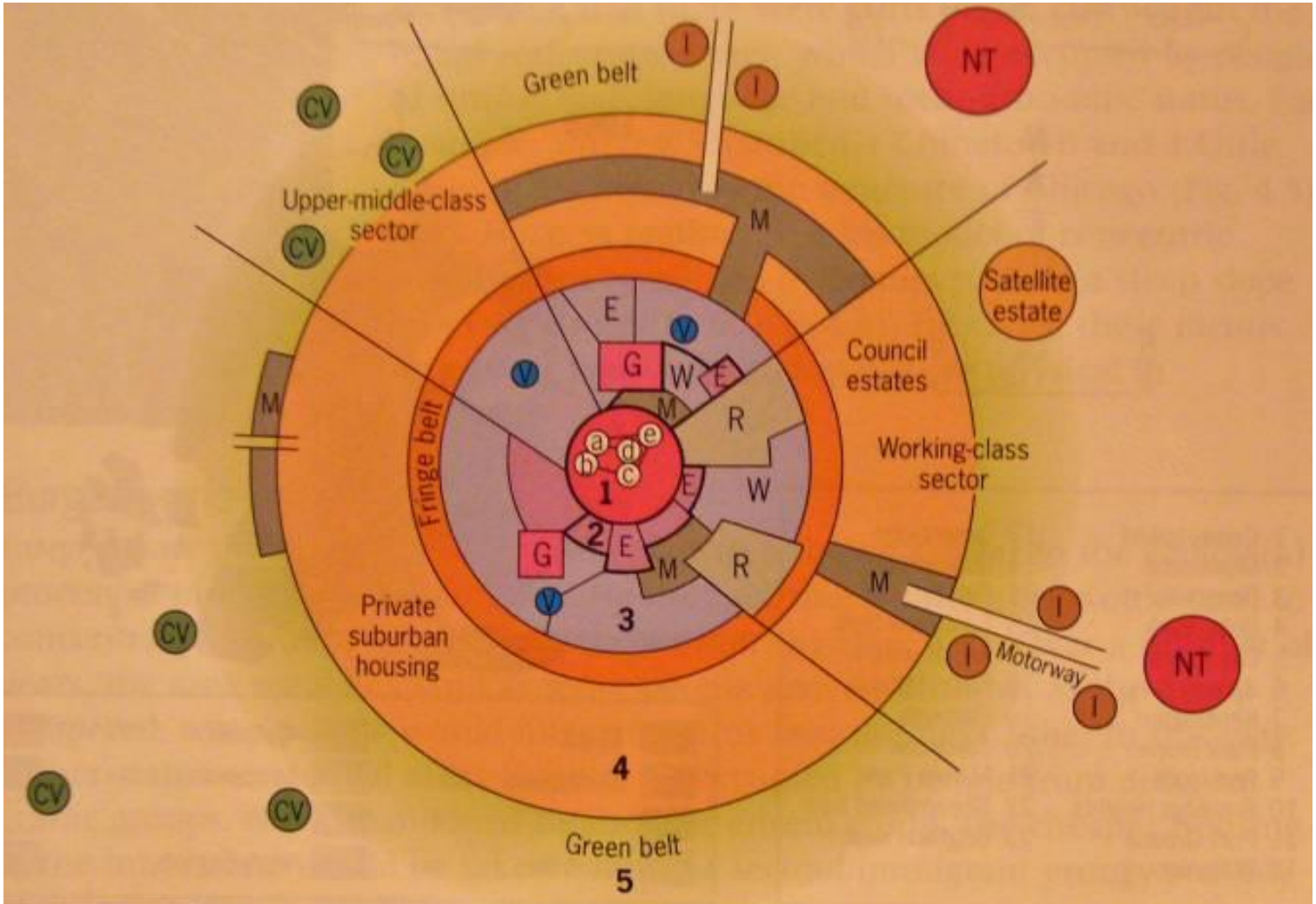
In recent years, geographers have begun to re-evaluate Burgess's work, and have tried to update it. One of the most important of the new models is that produced by Kearsley in 1983 (Fig. 4.10). Kearsley tried to adapt the Burgess model to include modern aspects of urban change such as inner city decline and decentralisation (the movement of people, shops and offices from the city centre to edge-of-city locations).



a–e Specialised nodes of activity within the CBD
CV Commuter village
E Ethnic-minority residential area
G Area of gentrification
I Inter-urban commercial/industrial site
M Manufacturing district located on radial and concentric routes or in twilight zone
R Local development scheme
V Encapsulated village
W Area of stable working-class communities

The divide between the inner city and the outer suburbs (often marked by a fringe belt of institutional land uses)
1 Central business zone
2 Transition zone
3 Pre-1918 residential development
4 Post-1918 suburban development
5 Ex-urban commuter zone
NT New Town

Kearsley's model includes contemporary urban processes and changes such as gentrification, commuter villages, development projects.



a–e Specialised nodes of activity within the CBD
CV Commuter village
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Hoyt – The Sector Model

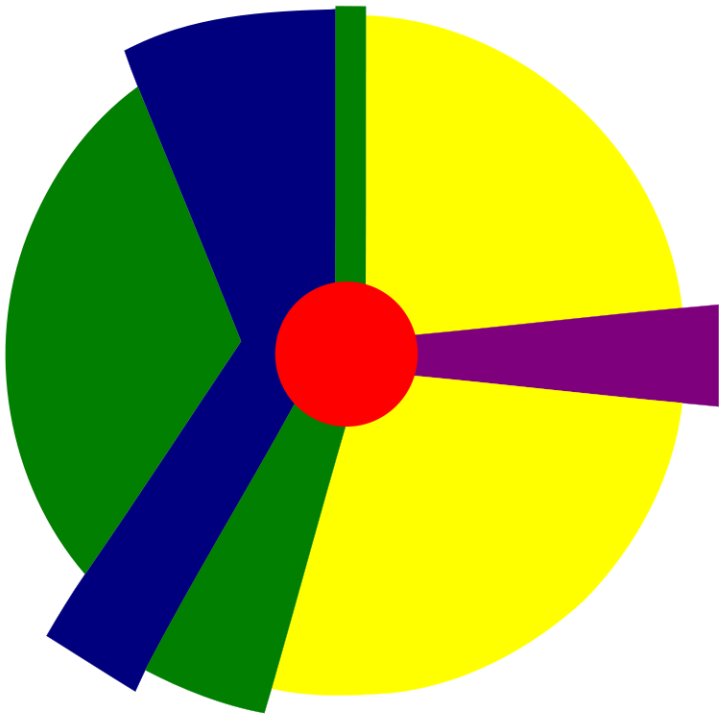
Created following a study of 142 cities in the USA

Main Emphasis

- **Access to major transport routes**
- **Explains the basic structure of a city**

The Sector Model

Homer Hoyt (1939)



Hoyt Sector Model Key

- CBD
- Factories/Industry
- Low class residential
- Middle class residential
- High class residential

Hoyt – The Sector Model

Sector Model compliments the Burgess model as it considers additional factor

Direction

- Land use is not a random distribution
- Land use is a function of Direction as well as distance
- In a given direction from center, a certain type of land use will prevail
- Land use mosaic consists of a series of sectors radiating out from city center

Hoyt – The Sector Model

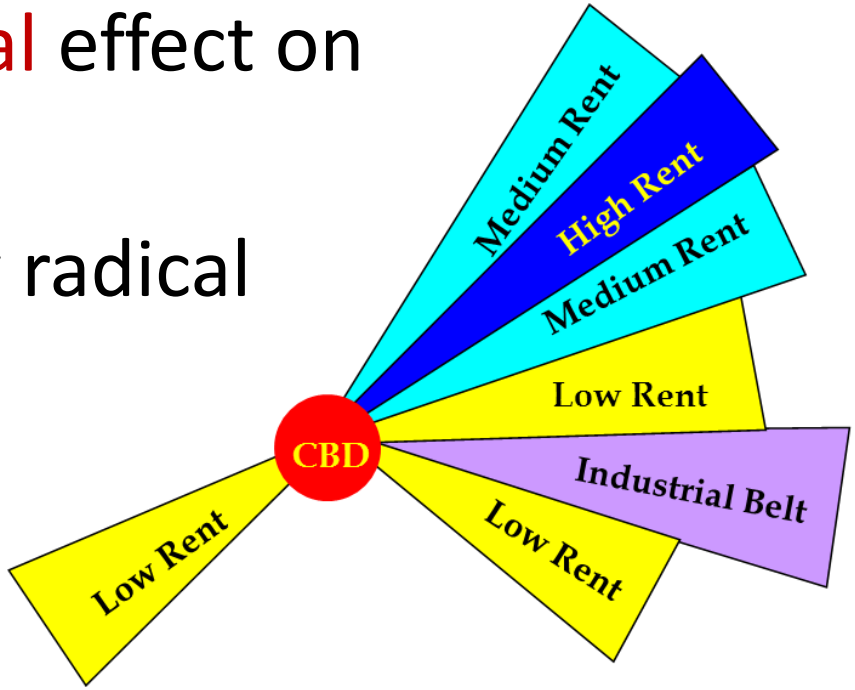
- Hoyt proposed the idea that towns grew as sectors or wedges.

"Different land use zones will develop along radial transport route from city core in sectoal forms, not concentric circles"

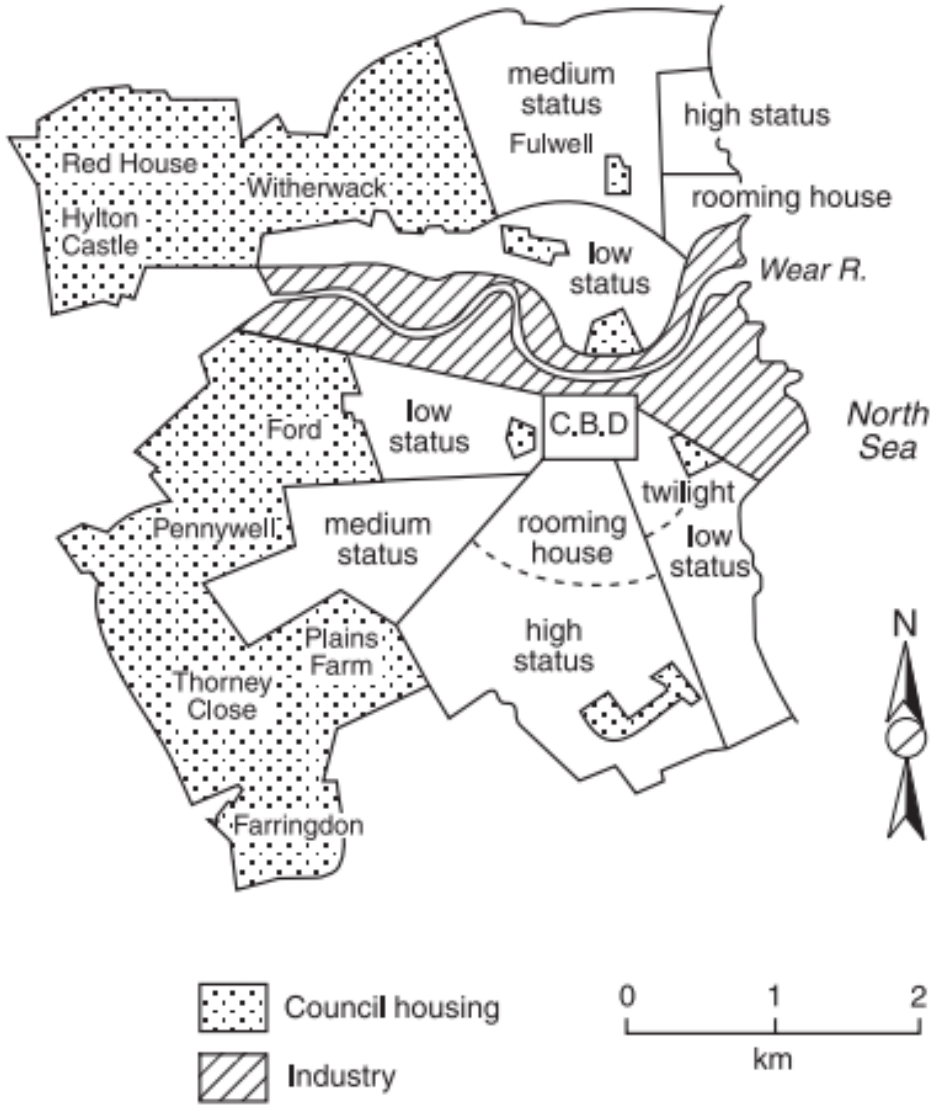
- Accessibility to the town or city centre influences the location of the functional zones/sectors.

Hoyt – The Sector Model

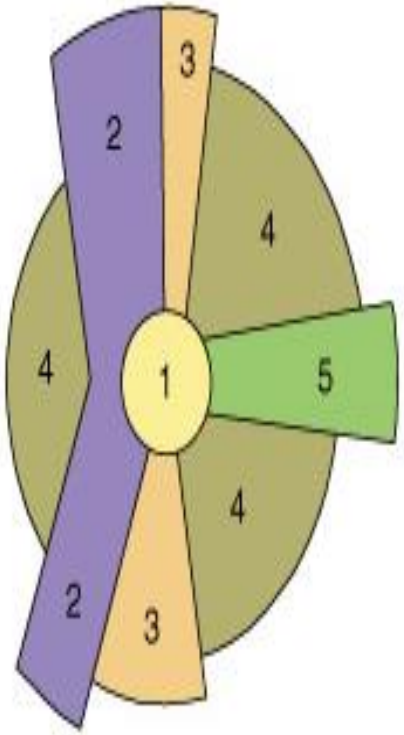
- Communication routes are mainly responsible for the creation of sectors
- transport has directional effect on land uses
- Rent varies primarily by radical sectors



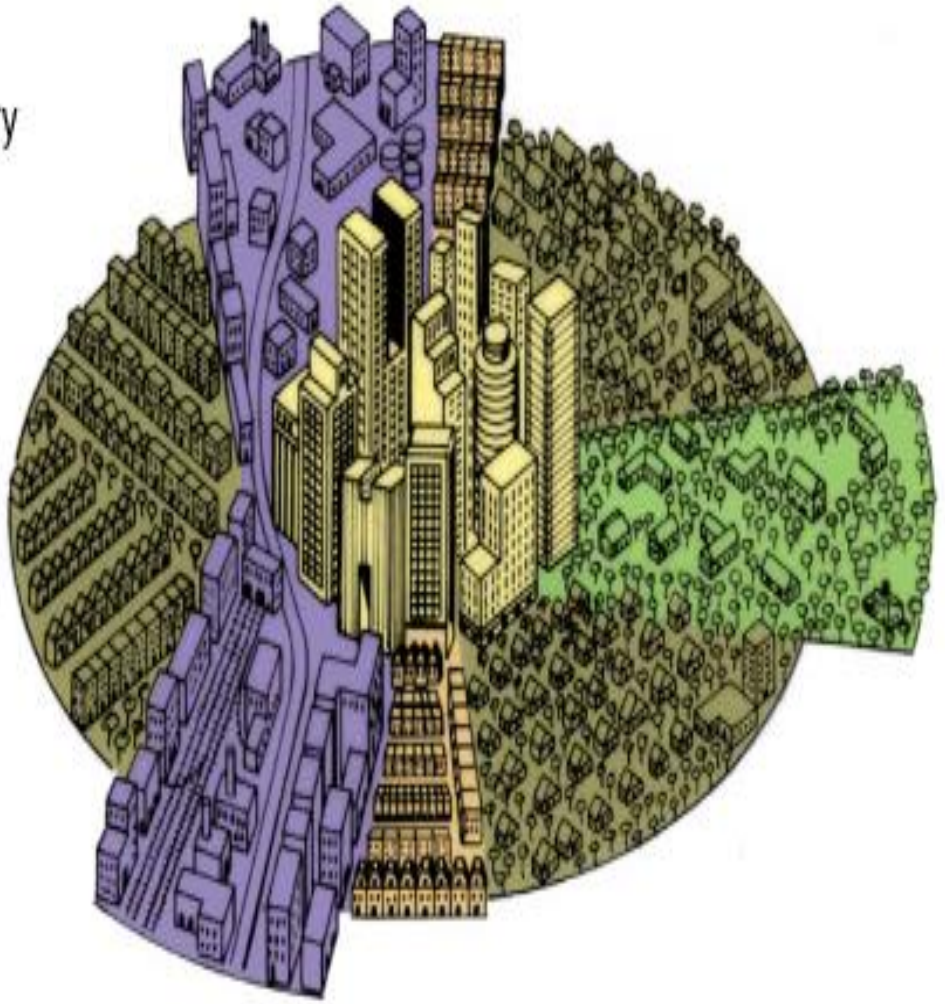
Hoyt's sector model application in Sunderland



Hoyt – The Sector Model



1. Central business district
2. Transportation and industry
3. Low-class residential
4. Middle-class residential
5. High-class residential



Stresses the importance of transportation corridors. Sees growth of various urban activities as expanding along roads, rivers, or train routes.

Hoyt – The Sector Model

CBD

>>This area contains shops, offices, banks, etc.

>>Land is expensive, and this area has high rents and multi storey buildings as a consequence.

>>There is very little space and competition is high.

>>Congestion levels are high.

>>Vegetated areas are sparse

Transportation and Industry

>>This contains old industries that have been positioned in this zone for many years.

>>Over the past 20 years, this industry has been non-profitable and has closed.

>>This has led to high unemployment

>>This area should be re-developed into modern business districts.

Low class residential housing

>>This group is made up of old 19th century buildings. With no gardens.

>>Often referred to as 'slums'. These were the buildings that were knocked down in the 1960's to be replaced with high rise flats.

>>The poorest people in the settlement live here, but is now highly admired and the place to live now (due to

Middle class residential housing

>>Semi-detached housing is found here with no gardens.

>>These houses are built on large estates.

>>Less expensive private estates can also be found here.

>>Often described as 'medium class residential' or 'inter-war' areas

High class residential housing

Commuter zone

>>This is a high class residential area where private, quality housing can be found.

>>Detached and semi-detached housing can be built on cheaper land.

>>Often lots of garages, big gardens, and many out buildings can be found here.

>>Most people commute to work from this zone as it is furthest away from the centre

Countryside Areas

>>In the countryside surrounding the urban area, those seeking to escape from the urban area can live in pleasant surroundings.

However, they are still close to work.

>>Many satellite villages and towns surround major urban areas allowing people to live further away from the main settlement.

Hoyt – The Sector Model

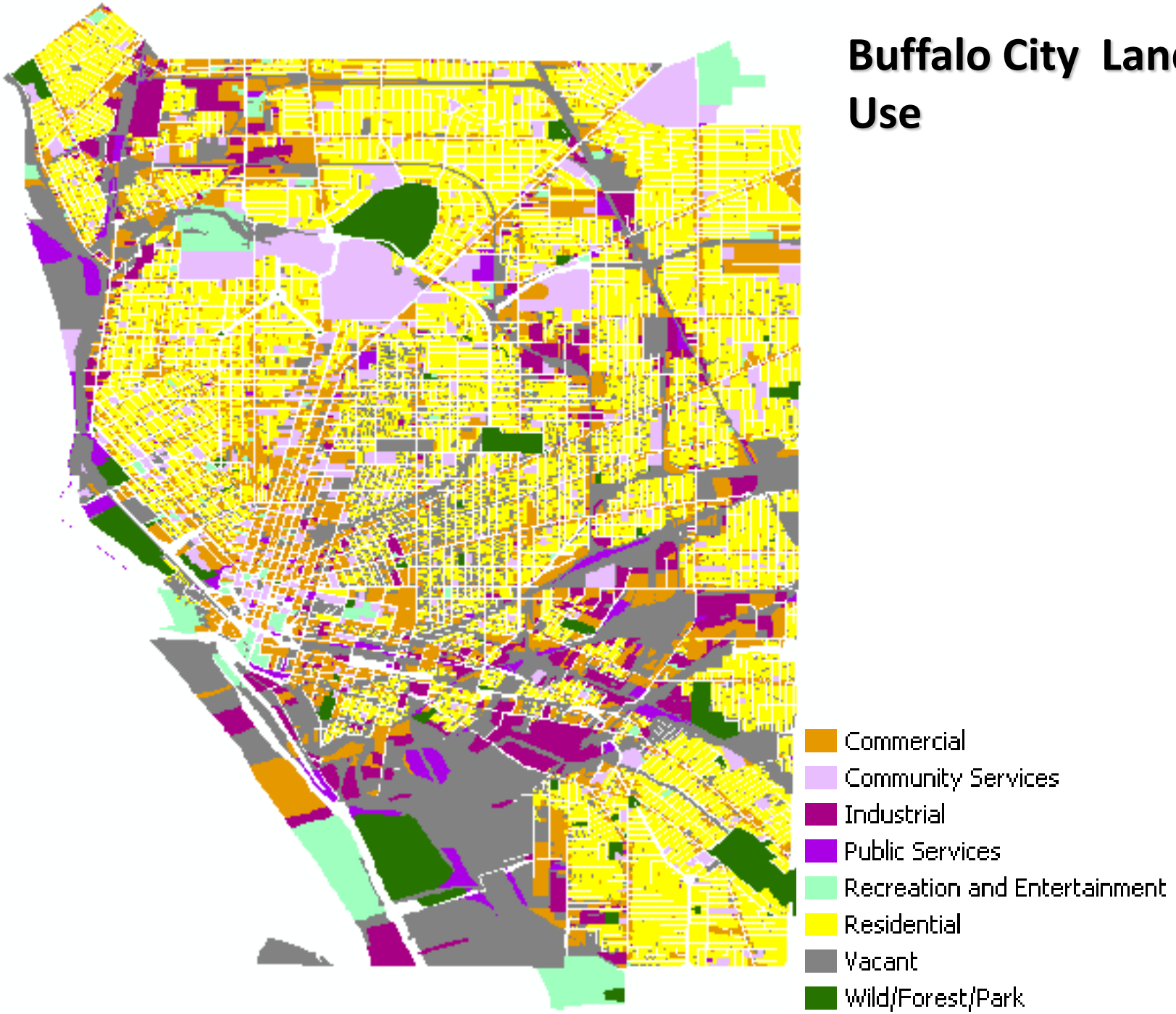
Basic Assumptions

- Wealthy people choose to live near transport links.
- They can afford cars so can live further from industry.
- Similar land uses attract similar land uses. This means certain functions are located in certain areas.
- But certain functions also repel each other.

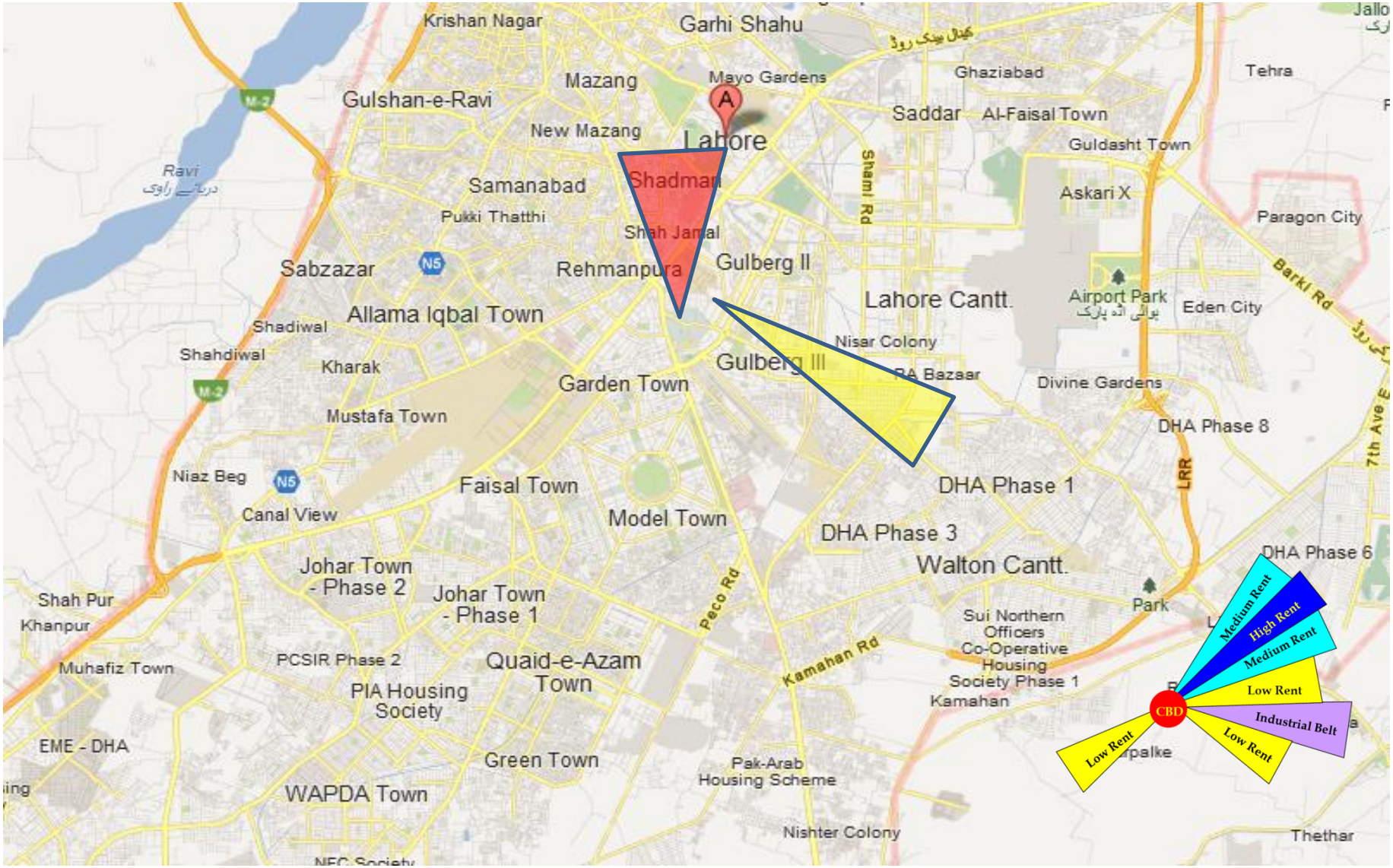
Criticisms

- Areas of low cost housing do not occur beside main roads as predicted.
- Assumes no planning controls.
- Does not take account of suburbs.

Buffalo City Land Use



Hoyt – The Sector Model

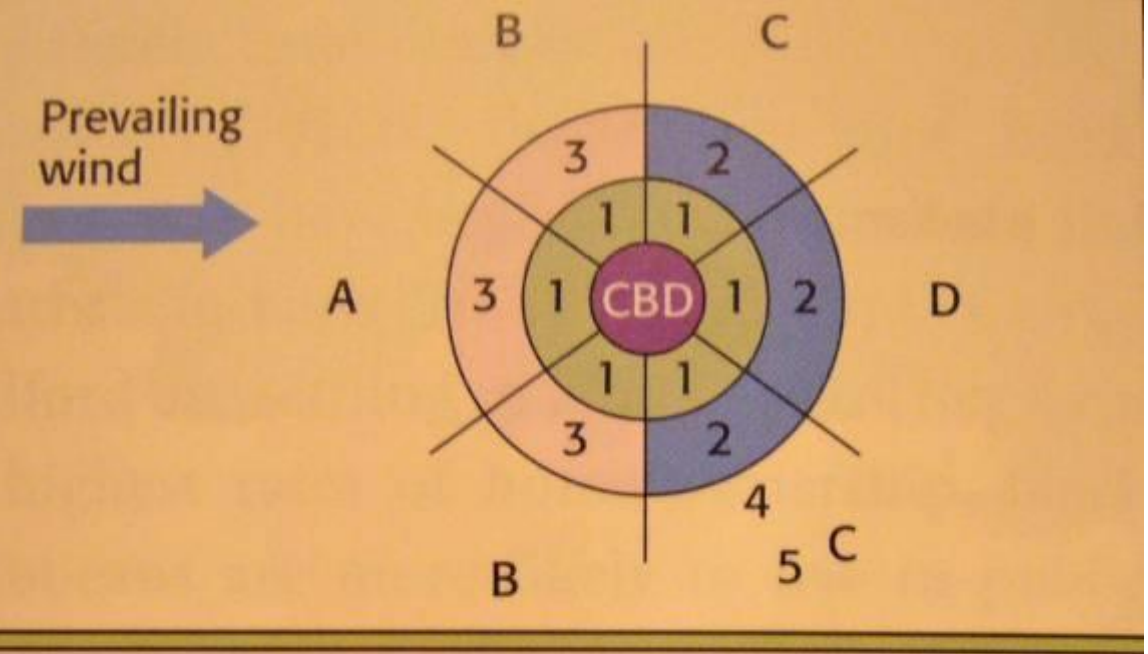


Sector model.

Shortcomings

- Applies well to some towns only
- Low cost housing is near industry and transportation proving Hoyt's model
- Theory based on 20th century and does not take into account cars which make commerce easier
- With cars, people can live anywhere and further from the city

Mann's concentric circles and sector model



Zones

1 Zone in transition

2 Small terraces

3 Late Victorian and larger old houses

4 Post-1918 residential, with post-1945 development mainly on the periphery

5 Commuter villages

Sectors

A Middle class

B Lower-middle class

C Working-class, public housing

D Industry and lower working-class

The industrial city of northern England

Devised by: Philip Mann, 1965

Main processes: proximity, centrality and accessibility

Given that the prevailing wind blows from the west, Mann assumed that industry would be in the east, and high-status housing upwind of the factories in the west. His model combines concentric zones and sectors, and also recognises the role played by local government in building public housing in the UK.


THANK
YOU!